

INSTITUTE OF ACTUARIES OF INDIA
EXAMINATIONS

May 2025

SA3 - General Insurance

Time allowed: 3 Hours 15 Minutes

Total Marks: 100

- Q. 1)** A new general insurance company in India that has just obtained Certificate of Registration is preparing to commence business with the Health and Crop segments. The company has an initial capital base of INR 100 Crores.

- i) What are the key regulatory considerations and requirements (by IRDAI) the Company would need to adhere to sell its first policy? (10)

The Board has recommended to allocate capital using an approach that captures tail risk.

The insurer ran 1,000 simulations of the returns on premium each line of business would generate, as well as the returns on investment a government security would generate. Below are the last 10 iterations of the simulation, sorted in ascending order (from highest to lowest returns):

Iteration	Health	Crop	Government Security
991	-50%	0%	7%
992	-50%	0%	7%
993	-50%	0%	7%
994	-50%	0%	7%
995	-50%	0%	7%
996	-100%	0%	7%
997	-100%	-100%	7%
998	-100%	-100%	7%
999	-100%	-200%	7%
1000	-100%	-200%	7%

	Health	Crop	Government Security
E(r)	15%	20%	7%
SD	20%	33%	0%

$\text{Cov}(R_{\text{health}}, R_{\text{crop}}) = 5\%$

- ii) Justify a risk measure that uses all of the iterations given above and determine which line of business is a more efficient use of capital from a risk-adjusted return on capital perspective. (5)
- iii) Identify and briefly describe at least two arguments for why management should not invest solely in the more efficient line of business as ascertained in Q1(ii) above. (4)

The company plans to write INR 100 crores of gross written premium in its first year of operation. By applying principles of mean-variance portfolio theory and portfolio optimisation, the company has determined that INR 64.8 crore and INR 35.2 crore are optimum premium levels for Health and Crop businesses respectively.

- iv) Identify and state briefly the importance of at least 5 Key Performance Indicators (KPIs) which can be used to effectively review these optimum premium levels for Health and Crop businesses after the company commences operations? (5)
- v) The board of directors want to compare the combined return and volatility of the two lines of business with government security and equities with expected return of 12% with a standard deviation of 30%. Use Sharpe ratio, which is defined as ratio of risk-adjusted return of an investment to its volatility, to assess efficiency of return relative to risk. (8)

The Board was particularly interested in the last 4 worst scenarios for Crop which was due to catastrophe. In this respect:

- vi) What is the basic structure of catastrophe model with respect to crop that can be built? (5)

vii) How can the catastrophe risk exposure be minimized in Crop insurance? (5)

viii) As an actuary at the Company, the Board has asked you to describe the factors that should be considered while deciding its investment policy considering the two lines of business. (9)

[51 Marks]

Q. 2) You are a reserving actuary of an Indian general insurance company. It writes industrial property business with high deductibles. The business is subject to large losses, which are usually reported quickly. Reserves are always set up in respect of these losses, but these are often found to be excessive or inadequate. You are concerned about the estimation of ultimate claims from this line of business, which has been carried out using the chain ladder method and has often been unstable in the past. It has been suggested to use Bornhuetter-Ferguson (BF) method to overcome the issue of unstable estimates.

i) By listing key claims characteristics of this book, state the possible reasons of Chain ladder method producing unstable reserve estimates? (5)

ii) Justify the suggestion of using BF method. (3)

iii) What will be your recommendation and next steps for implementing this change in method. (4)

It was also discovered that the company has had changes in internal environment which could have distorted the paid and reported development patterns.

iv) Identify 5 such changes that could distort the development patterns. (5)

v) For each of the changes identified above, briefly describe how will it be diagnosed and the likely observations suggesting the change. (10)

The company has the following two reinsurance treaties:

- 4-line surplus treaty with maximum retention of INR 500,00,000
- INR 200,00,000 xs INR 500,00,000 individual risk excess of loss

It also reinsures some risks on facultative basis.

vi) Describe briefly the main features of surplus, individual excess of loss and facultative reinsurance using the treaties above. (6)

vii) Discuss briefly the features of an individual risk that the company should consider when deciding how much of that risk it should cede to the reinsurer. (6)

[39 Marks]

Q. 3) You are a pricing actuary at an insurance company primarily writing motor and health insurance.

A health employer-employee policy for a large insured, YokoTech is expiring on July 1, 2025. The following changes have taken place at YokoTech since the policy was last issued on July 1, 2024:

- YokoTech has implemented wellness and various health programs with incentives to reduce the expected number of claims.
- YokoTech has doubled the number of their employees.

i) Briefly describe how each of these changes could be accounted for in the final premium for the July 1, 2025 renewal policy. (4)

With the rise of telematics and usage-based insurance, the company is considering changing its exposure base from car-years to hours driven.

ii) Evaluate hours driven as exposure base from actuarial and operational perspective. (6)

[10 Marks]
